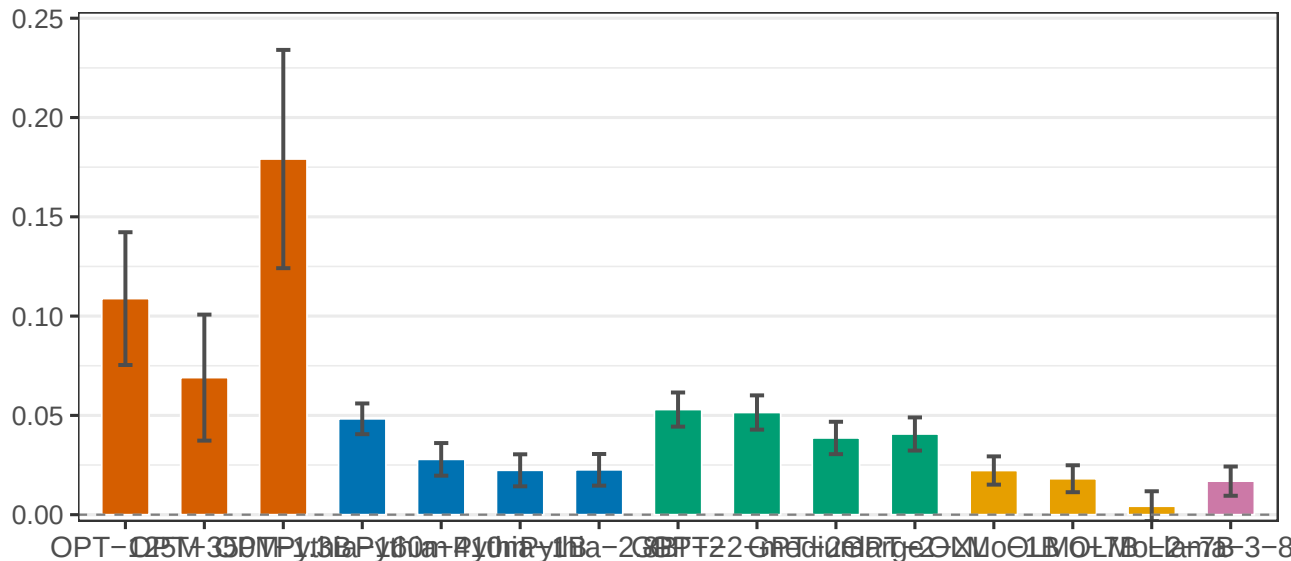


Context-dependence (memorisation via attention) at the final layer

Larger delta = more memorisation encoded in cross-word attention

BabyLM (2B tok) GPT-2 (WebText) Llama 3 (Meta) OLMo (Dolma) Pythia (300B)



Corpus-frequency regression: $\text{Im}(|\text{residual}| \sim \log_{\text{freq}})$ at the final layer, mean_pooled mode.

BabyLM: BabyLM-corpus frequencies; Pythia: Pile frequencies.

Error bars: 95% CI propagated from individual β SEs.

$\Delta = \text{default } \beta - \text{attn_zeroed } \beta$ (mean_pooled, final layer)